

11 For the reaction $\text{NH}_4\text{NO}_3(\text{s}) \rightarrow \text{N}_2\text{O}(\text{g}) + 2\text{H}_2\text{O}(\text{l})$, $\Delta H^\circ = -124 \text{ kJ mol}^{-1}$.

Which statement is correct?

- A** The reaction is spontaneous only at low temperature.
- B** The reaction is spontaneous only at high temperature.
- C** The reaction is not spontaneous at any temperature.
- D** The reaction is spontaneous at all temperatures.

12 Hydrogen peroxide is a liquid at room temperature. It decomposes to form water and oxygen gas.